

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.1: The Periodic Table — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 2.1: The Periodic Table
Driving Question	DRIVING QUESTION: Why do the fertiliser bags on Kenyan farms carry such specific and different chemical formulae — and what does the inner structure of an atom have to do with it?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring the Phenomenon and Reviewing Atomic Structure	Two fertiliser bags used on Kenyan farms — CAN ($\text{Ca}(\text{NO}_3)_2 \text{NH}_4\text{NO}_3$) and DAP ($(\text{NH}_4)_2\text{HPO}_4$) — look similar on the outside as white crystalline solids that dissolve in water, yet they carry completely different chemical formulae and affect plant growth differently. Both contain nitrogen, but their formulae differ in the elements present and the ratios in which those elements appear.	Atoms of each element have a fixed number of protons (atomic number) and a characteristic electron arrangement described using the 2-8-8 shell model. The elements in the fertiliser formulae — H, N, O, Ca, P — each have a specific atomic number and a specific pattern of electrons distributed across their shells. The number of electrons in the outermost shell appears to be particularly important.	We cannot yet fully explain why the formulae are different, but we now know that the answer is connected to the electron arrangements of the atoms involved. The number of outer-shell electrons of calcium (2.8.8.2 — two outer electrons) somehow relates to the 2 appearing in $\text{Ca}(\text{NO}_3)_2$. Understanding this connection between electron arrangement and formula will be built across the sub-strand.
Lesson 2 Electron Configuration and the Periodic Law: Finding the Hidden Pattern	Learners sorted 20 element cards by electron arrangement data and discovered that the number of outer-shell (valence) electrons repeats in a predictable pattern as atomic number increases — elements with 1 outer electron appear at regular intervals, as do elements with 2, 3, and so on up to 8 (noble gases).	The repeating pattern of valence electrons is called the periodic law. Elements are arranged in the periodic table in rows called periods (which correspond to the number of occupied electron shells) and columns called groups (which correspond to the number of valence electrons). Calcium (in CAN) is in Group 2 because it has 2 valence electrons; nitrogen (in both CAN and DAP) is in Group 15 because it has 5 valence electrons.	The periodic law begins to explain why CAN and DAP have different formulae: the number of valence electrons determines how many electrons an atom can lose or gain to become stable, which in turn determines the charge of the ion it forms, which determines the ratio in which elements combine in a compound. Calcium's 2 valence electrons mean it forms Ca^{2+} — a fact that will directly explain the '2' in $\text{Ca}(\text{NO}_3)_2$ once we study ion formation in Lesson 3.
Lesson 3 Stability, Electron	Learners observed three pieces of evidence: (1) noble gases (He, Ne, Ar) all	Atoms are most stable when their outer shell is full — matching a noble gas	The specific ion charges of Ca^{2+} and N^{3-} are not arbitrary — they are determined by

Affinity, and the Noble Gas Rule	have full outer electron shells; (2) when the first 20 elements are sorted by outer electron count, metals cluster on the low side (1–3) and reactive non-metals on the high side (5–7); (3) dot-and-cross diagrams show that Na loses 1 electron to achieve neon's configuration (Na^+) and Cl gains 1 electron to achieve argon's configuration (Cl^-).	configuration. Atoms with few outer electrons (metals) lose electrons and form positively charged cations. Atoms with nearly full outer shells (non-metals) gain electrons and form negatively charged anions. Calcium (2,8,8,2) loses 2 electrons to achieve argon's configuration (2,8,8), forming Ca^{2+}. Nitrogen (2,5) gains 3 electrons to achieve neon's configuration (2,8), forming N^{3-}. Electron affinity describes an atom's tendency to attract electrons.	how many electrons each atom must lose or gain to achieve the nearest noble gas electron configuration. This is why calcium always forms a 2+ ion (never 1+ or 3+) and nitrogen always forms a 3– ion. These fixed charges will directly determine the combining ratio of calcium and nitrogen ions in compounds, explaining why the CAN fertiliser formula is $\text{Ca}(\text{NO}_3)_2$ and not some other ratio.
Lesson 4 Ion Formation: Dot-and-Cross Diagrams	Learners watched an animation/simulation of ion formation for elements from the first 20 of the periodic table (Na, Mg, Al, Cl, O, Ca) and recorded in an observation table: electron arrangement before and after, number of electrons lost or gained, the ion formed, and the resulting charge.	Atoms of metallic elements in Groups 1, 2, and 3 lose electrons equal to their group number to form positively charged ions (cations: Na^+, Mg^{2+}, Al^{3+}, Ca^{2+}); atoms of non-metallic elements in Groups 6 and 7 gain electrons to complete their outer shell and form negatively charged ions (anions: O^{2-}, Cl^-). The ionic charge is determined by the atom's electron arrangement and its position in the periodic table — the same inner-atomic structure that organises the periodic table.	The specific formulae on the CAN ($\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$) and DAP ($(\text{NH}_4)_2\text{HPO}_4$) fertiliser bags used on the Kericho tea farm are not arbitrary. Calcium, in Group 2, always forms Ca^{2+} by losing its two outer electrons to achieve a stable noble-gas configuration. Nitrogen and oxygen form ions whose charges are likewise fixed by their electron arrangements. These fixed ionic charges mean that elements must combine in specific ratios to produce electrically neutral compounds — which is why the formula is $\text{Ca}(\text{NO}_3)_2$ and not CaNO_3. The 'hidden pattern' the learner noticed on the fertiliser bag is the direct consequence of ion formation driven by electron arrangement.
Lesson 5 Valency, Oxidation Numbers, and Radicals: Unlocking the Hidden Pattern in Fertiliser Formulae	Learners completed a Valency Derivation Table for the first 20 elements, deriving ion charges and valencies from their Lesson 3–4 electron configuration and ion formation notes. They sorted seven KICD Radical Cards (NH_4^+, SO_4^{2-}, NO_3^-, OH^-, CO_3^{2-}, PO_4^{3-}, HCO_3^-) by charge and matched them to the CAN and DAP formulae from the Kericho phenomenon.	Valency is the combining power of an element or radical, determined by the number of electrons lost, gained, or shared; it equals the magnitude of the ion charge. Oxidation numbers formalise the charge carried by an atom in a compound. The criss-cross rule uses valencies to generate correct chemical formulae. The seven KICD radicals are polyatomic ions that remain intact in reactions. $\text{Ca}(\text{NO}_3)_2$ has subscript 2 on nitrate because calcium's valency (2) criss-crosses with nitrate's valency (1); $(\text{NH}_4)_2\text{HPO}_4$ has subscript 2 on ammonium because hydrogen phosphate's (HPO_4^{2-}) valency (2) criss-crosses with ammonium's valency (1).	The specific subscript numbers in fertiliser formulae like CAN and DAP are not arbitrary — they are mathematical consequences of the valencies (ion charges) of the component elements and radicals, which are themselves consequences of electron arrangement in the atom. This is why a farmer in Kericho sees $\text{Ca}(\text{NO}_3)_2$ and not CaNO_3: calcium must combine with exactly two nitrate units to achieve electrical neutrality, because calcium carries a 2+ charge and each nitrate carries a 1– charge.

Lesson 6 Writing Chemical Formulae of Common Compounds	Learners observed a teacher-modelled demonstration of the four-step crossover-valency method applied to four progressively complex compounds (NaCl, MgO, Al₂O₃, CaCO₃), including the bracket rule for polyatomic radicals, and read a short structured text explaining when brackets must be used in ionic formulae.	Learners can apply the crossover-valency method to write and verify chemical formulae for common compounds including binary compounds and those containing radicals; they can decode the fertiliser formulae CAN [Ca(NO₃)₂·NH₄NO₃] and DAP [(NH₄)₂HPO₄] from the anchoring phenomenon, and explain why the two formulae differ in terms of the valency and charge-balance requirements of their constituent ions.	The fixed ratios in chemical formulae are not arbitrary — they are determined by the valencies of the combining ions, which arise from electron arrangement, which reflects atomic structure. The crossover-valency method encodes nature's requirement that every compound must be electrically neutral (total positive charge = total negative charge). This is why the learner in Kericho sees different formulae on her two fertiliser bags: HPO₄²⁻ has a valency of 2 and therefore demands two NH₄⁺ ions, while NO₃⁻ has a valency of 1 and requires only one NH₄⁺ — the hidden pattern is charge balance, rooted in atomic structure.
Lesson 7 Balancing Chemical Equations: Writing the Language of Fertiliser Formulae	Learners observed the teacher model a 4-step balancing strategy (write → count → balance by inspection → verify) applied live to Na + Cl₂ → NaCl and Mg + O₂ → MgO, using a maize-lorry analogy to distinguish coefficients from subscripts. They also observed four unbalanced equation cards in a gallery walk, noting atom counts for NaCl, MgO, Ca(NO₃)₂, and (NH₄)₂HPO₄.	Learners learned that: (i) a balanced chemical equation has the same number of each type of atom on both sides; (ii) only coefficients (not subscripts) may be changed when balancing; (iii) the 4-step strategy (write → count → balance → verify) can be applied to equations involving elements and compounds of the first 20 elements; (iv) balanced equations prove that atoms combine in fixed ratios determined by valency — which is exactly why the fertiliser formulae CAN [Ca(NO₃)₂·NH₄NO₃] and DAP [(NH₄)₂HPO₄] carry those specific and unalterable chemical formulae.	Learners explained — through paired practice, peer checking, DQB annotations, and cumulative model sentence stems — that the fixed ratios in fertiliser formulae are not arbitrary: they are a direct consequence of the valencies of the constituent elements and polyatomic ions, and these valencies arise from the electron arrangements of atoms (established in Lessons 2–4). A balanced equation is the formal, conservation-law-backed proof of those ratios. The Kericho learner's puzzle is therefore answered: the inner structure of the atom (electron configuration → valency) determines what formula a compound gets, and balancing the equation confirms that the ratio is the only one consistent with the law of conservation of mass.
Lesson 8 Model Building, Project Showcase and Driving Question Board Resolution: From Electron Arrangement to Fertiliser Formula	Learners observed electron-configuration models built from bottle tops, wire, and clay representing the first 20 elements; compound-formula charts showing names, formulae, and valency justifications for Kenyan compounds including CAN and DAP; and the completed Driving Question Board with sticky-note predictions from Lesson 1 alongside final two-sentence resolutions.	Learners established that the inner structure of an atom — specifically the number of electrons in the outer shell — determines how many electrons an atom loses, gains, or shares, producing a specific ion charge called valency; that valency values are cross-multiplied to produce the only chemically correct formula for a compound; and that the formulae Ca(NO₃)₂ · NH₄NO₃ for CAN and (NH₄)₂HPO₄ for DAP are therefore not arbitrary labels but necessary consequences of the electron arrangements of calcium, nitrogen,	Learners explained that: a tea farmer in Kericho sees different formulae on her fertiliser bags because calcium (Group 2, arrangement 2,8,8,2) always forms Ca²⁺ while ammonium carries a 1+ charge and nitrate a 1- charge and hydrogen phosphate (HPO₄²⁻, in DAP) a 2- charge; cross-multiplying these valencies produces the fixed, unalterable ratios encoded in the formulae; balanced equations verify that no atoms are created or destroyed in these fixed-ratio combinations; and therefore the specific formula on each bag is a direct,

		hydrogen, oxygen, and phosphorus.	readable record of the electron structure of the atoms inside — making the periodic table the key to understanding Kenyan agriculture at the molecular level.
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ATTRIBUTION AND LICENSING

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